

Agent-oriented modality in the languages of North America

Jessica K. Ivania^a, Alexandra Nogina^b, Taras Zakharko^a

^aUniversity of California, Santa Barbara

^bThe Hebrew University of Jerusalem

Abstract

This study examines the morphosyntactic and functional encoding of agent-oriented modality in a maximally diverse sample of languages from North America. Focusing on necessity, obligation, ability, and desire, we show that agent-oriented modality is expressed through a wide range of construction types, including tense-aspect-mood morphology, modal verbs, non-verbal formatives, and complex clauses. We show that clustering the constructions by their formal properties yields three groups that align closely with the modal functions of necessity, ability, and desire. We further show that the constructions encoding different modal functions tend to differ in their morphosyntactic profiles, with strong necessity and obligation in particular showing systematic associations with tense marking, and weak and strong necessity with mood marking, respectively. These findings suggest that agent-oriented modality is not a uniform grammatical category, but a structured functional domain whose subtypes recruit material from different areas of grammar. The study contributes to a typological understanding of modality by combining constructional detail and an autotypologizing approach to cross-linguistic comparison.*

Keywords: modality; quantitative typology; autotypology; morphosyntax; North-American languages

1 Introduction

In this study, we explore the domain of modality, with a focus on agent-oriented modality, in the languages of North America. While modality has been extensively discussed in formal semantic and pragmatic terms, especially following work by Kratzer

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(1981, 2012) and Portner (2009, 2018), we lack a systematic overview of how this domain is morphosyntactically encoded in the languages of the world, the functions and related subtypes that we find, and the associations between construction types and modal functions. In particular, it is unclear from which domains modal propositions recruit morphosyntactic material for their expression, including tense, aspect, mood, modal verbs, and non-verbal formatives.

To address this gap, we explore the morphosyntactic and functional encoding of agent-oriented modality. By agent-oriented modality, we mean the morphosyntactic expression of necessity and obligation, desire, and ability, following Bybee et al. (1994)'s terminology. We focus on the languages of North America, using a maximally diverse sample. Methodologically, we collect and annotate these forms using the autotypologizing method to capture fine-grained variation.

We show quantitatively that, in our sample, clustering the constructions based on their formal properties yields three groups that closely align with the expression of three main modal functions: necessity (including obligation, and strong and weak necessity), ability, and desire. We also show that certain modal functions and related subtypes, such as necessity, are systematically associated with particular morphosyntactic realizations, especially tense and mood markers. These findings contribute to a broader understanding of how modal meanings may arise from, and interact with, other grammatical domains in the languages of the world.

The paper is structured as follows. Section 2 introduces agent-oriented modality. Section 3 presents the data and the sample. Section 4 illustrates the morphosyntactic (4.1) and semantic (4.2) variation found in the data. Section 5 presents the data structuring and autotypologizing method. Section 6 illustrates the main findings (6.1-6.3), including qualitative observations on less common structures found in the data (6.4). Section 7 concludes with a summary.

2 Agent-oriented modality

The term 'agent-oriented modality' (AOM) as used in this article, follows the framework proposed by Bybee et al. (1994). Agent-oriented modality concerns the existence of internal or external conditions on an agent with respect to the performance of the action expressed by the main predicate. In other words, AOM does not primarily evaluate the truth of a proposition, but rather specifies whether, and under what conditions, an agent is able, required, inclined, or compelled to carry out an event.

Within this domain, Bybee et al. distinguish four major categories: obligation, necessity, ability, and desire. These categories differ according to the type of conditions imposed on the agent. Obligation involves external, typically social or normative conditions, as in *They must leave now*, where the agent is required to act by some authority, rule, or expectations. Necessity involves external or physical conditions, as in

I have to stop because the road is blocked, where the action is required by circumstances rather than by social obligations. Ability involves internal enabling conditions, as in *They can swim*, where the relevant condition concerns the agent's capacity. Desire involves internal volitional conditions, as in *I want to go*, where the agent is presented as having an inclination towards an event.

Different types of agent-oriented modality are often associated with different lexical sources. Markers of obligation and necessity frequently develop from lexical items directly related to obligation or need, such as verbs meaning 'want' or 'need'. They may also arise from more general verb-like elements, including forms meaning 'be', 'become', or 'have'. In the domain of ability, a common lexical source is the verb 'know', especially in the sense 'know how to'. Ability meanings may also develop from expressions of physical capacity, which then expand to more general notions of ability. In addition, dynamic or telic verbs such as 'arrive' or 'get' may develop into markers of ability, possibility, or successful completion.

The four major types of AOM may also undergo diachronic extension, giving rise to additional meanings. Necessity may develop into desire, while markers of desire may further extend to meanings of intention or willingness. Ability markers may expand toward more general root possibility. More broadly, agent-oriented modality is also an important diachronic source for other modal domains. In many languages, markers that originally express agent-oriented meanings later develop epistemic uses, shifting from conditions on an agent's action to assessments of the likelihood or certainty of a proposition. Although the present study does not focus on epistemic modality, several cases of multifunctionality between agent-oriented and epistemic meanings are observed in the data.

3 Sample

Our language sample includes 20 North American languages and is designed to be maximally diverse typologically, featuring one language per family (with isolates treated as independent families). Within each family, language choice was motivated by convenience and bibliographical constraints, namely: (1) the availability of reference grammars and (2) the existence of an exhaustive or nearly exhaustive description of modality. The languages, their corresponding glottocodes, and their language families (as well as the respective descriptive sources) are presented in Table 1. Language names reflect Glottolog naming conventions.

Language	Language family	Glottoco	#Construction	Source(s)
Barbareño	Chumashan	barb1263	3	Wash 2001
Caddo	Caddoan	cadd1256	3	Melnar 1998, Chafe 2018
Central Alaskan Yupik	Eskimo-Aleut	cent2127	5	Miyaoka 2012

Creek	Muskogean	cree1270	4	Martin 2011
Halkomelem	Salishan	halk1245	4	Suttles 2004
Hidatsa	Siouan	hida1246	3	Park 2012
Huehuetla Tepehua	Totonacan	hueh1236	4	Kung 2007
Ixil	Mayan	ixil1251	4	Adell 2019
Kiowa	Kiowa-Tanoan	kiow1266	3	Watkins 1984
Meskwaki	Algic	mesk124	4	Dahlstrom ND
North Slavey	Athabaskan-Eyak-	nort2942	4	Rice 1989
Northeast Chiapas	Mixe-Zoque	rayo1235	2	Faarlund 2012
Nuu-chah-nulth	Wakashan	nuuc1236	4	Davidson 2002, Nakayama
Onondaga	Iroquoian	onon1246	7	Woodbury 2018
Seri	isolate	seri1257	5	Marlett 1981
Sochiapam Chinantec	Otomanguean	soch1239	5	Foris 2000
Southern Pomo	Pomoan	sout2984	2	Walker 2020
Tipai-Kumeyaay-	Cochimi-Yuman	kumi1248	3	Miller 2001
Ute-Southern Paiute	Uto-Aztecan	utes1238	5	Givón 2011
Yuchi	isolate	yuch1247	5	Linn 2001

Table 1: Languages included in the sample.

For each language in the sample, we collected and annotated constructions encoding the modal functions of necessity, ability, and desire. Methodologically, we focused on constructions explicitly identified in the source material as encoding these modal functions and illustrated with glossed examples. Once the primary forms for each modal flavor were established, we identified additional examples to serve as the basis for compiling the variation space of AOM within these languages. The number of constructions per language ranges from two to seven, with a total of 79 constructions identified across the North American sample. Figure 1 depicts the geographical distribution of these languages.

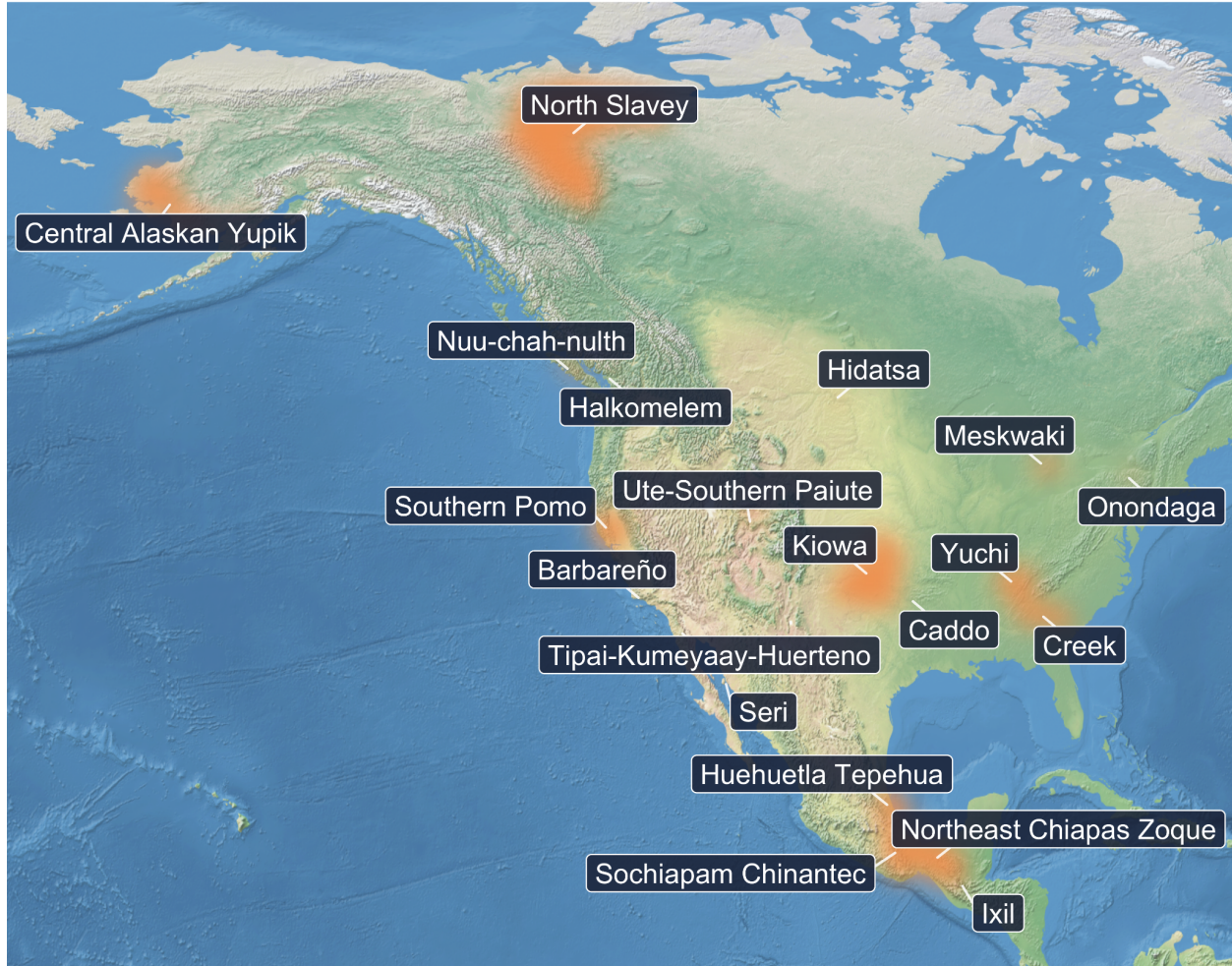


Figure 1: Geographical distribution of the languages in the sample

4 Variation in agent-oriented modality

The sample exhibits considerable variation in the expression of agent-oriented modality. This diversity encompasses both the formal encoding of modal functions within and across languages, and the semantic distribution of various modal subtypes identified in the data. In what follows, we provide illustrative examples of the morphosyntactic (Section 4.1) and semantic (Section 4.2) variation found in our study.

4.1 Morphosyntactic variation

The encoding of modal functions across the North American languages in the sample is characterized by notable morphosyntactic diversity. While Onondaga (Iroquoian, Example 1) utilizes the modal verb *á·gɛk* (‘should’) to convey both strong and weak necessity, other languages, such as Central Alaskan Yupik (Eskimo-Aleut, Example 2), utilize dedicated modal markers. In the latter, the suffix *-narq* is used specifically to express necessity and is suffixed to the main verb (e.g., *ner-* ‘eat’).

- (1) Onondaga (Iroquoian)
da nis á·gək ɛ-t-hs-ahgwehnɛhd-a?
 LNK PRON shouldFUT-CIS-2SG.A-descend-PNC
ɛ-hs-ade?gw-a?
 FUT-2SG.A-escape-PNC
 ‘So, as to you, you should come down [and] escape.’ (Woodbury 2018: 404)
- (2) Central Alaskan Yupik (Eskimo-Aleut)
ner-narq-aanga neq-mek
 eat-NEC-IND.3SG.1SG fish-ABM.SG
 ‘I have to eat fish.’ (Miyaoaka 2012: 1059)

Some languages employ tense and/or mood markers for the encoding of necessity. In Sochiapan Chinantec (Otomanguean; Example 3), necessity is expressed through the hortative marker *kui^H-*, which is prefixed to the verb. Hidatsa (Siouan; Example 4) shows a similar pattern, using the future marker *wihi* to express necessity.

- (3) Sochiapan Chinantec (Otomanguean)
kui^H-ŋji^{MH} hná^{HL} ?ŋju^{LM}
 HORT-go.home.IA.FUT.1SG I house.1SG
 ‘I should go home.’ (Foris 2000: 119)
- (4) Hidatsa (Siouan)
maa-há-waa-dhaa-wihi-?a-c
 3OBJ.PL-abandon-1CAUS.DIR-NEG-1PL.FUT-PL-DECL
 ‘We should ditch them.’ (Park 2012: 284)

Other languages use complex morphosyntactic strategies to encode necessity. Southern Paiute (Uto-Aztecan; Example 5) employs a combination of the irrealis marker *-vaa* and the nominalizer *-nā*, a construction that extends to subjunctive contexts (Givón 2011: 135). In Kiowa (Kiowa-Tanoan; Example 6), weak necessity involves the modal *dá* paired with the general extender *-àl*, with the main verb appearing in the imperative via the suffix *-bó*.

- (5) Ute-Southern Paiute (Uto-Aztecan)
mamachi ?uway wíʔka-vaa-nā-?u
 woman.GEN 3SG.GEN work-IRR-NMLZ-3SG

‘The woman must work.’ (Givón 2011: 140)

- (6) Kiowa (Kiowa-Tanoan)

<i>dá-àl</i>	<i>ám</i>	<i>yí:dè</i>	<i>kôl</i>	<i>pí:gyá</i>
must-also	you	both	some	food

gyát-bó

X.A>1PL.P>PL.OBJ-bring.IMP

‘You should also bring some food for us.’ (Watkins 1984: 222)

Similar variation is observed in other modal functions, such as ability. In Nuuchah-nulth (Wakashan; see Example 7), ability is encoded by the modal verb *?u?umhi* (‘be able’), which precedes the verb denoting the main event. Conversely, North Slavey (Athabaskan-Eyak-Tlingit; see Example 8) employs the dedicated modal marker *hoyji* to express this function.

- (7) Nuuchah-nulth (Wakashan)

<i>?u?umhi-matak-s</i>	<i>k^wis-<x>-aqs-ił</i>
be.able.to-probably-1SG	be.different-?-being.on.the.side-MOM

‘Maybe I can go across the strait.’ (Nakayama 2001: 115)

- (8) North Slavey (Athabaskan-Eyak-Tlingit)

<i>dene</i>	<i>hoyji</i>	<i>duyile</i>	<i>?eghálayeda</i>
person	any	can	3.work

‘Anyone can do that work.’ (Rice 1989: 406)

Other languages utilize mood markers or complex periphrastic structures for ability. In Seri (isolate; Example 9), the irrealis marker *tm-* is prefixed to the main verb. Ixil (Mayan; Example 10) presents a more complex structure: the modal verb *utch* (‘be possible’) is modified by the prospectual aspect clitic *la?*= and requires a phrasal argument marked by the subordinator *-t*.

- (9) Seri (isolate)

<i>?ant</i>	<i>po-fi:</i>	<i>ta</i>	<i>tm-a:?</i>	<i>χap</i>
land	IRR1-?	SR	IRR2-PASS-dig.up	

‘Tomorrow it can be dug up.’ (Marlett 1981: 23)

- (10) Ixil (Mayan)

<i>la?=utch</i>	<i>qu-txey-t</i>	<i>qu-txay</i>
PROS=be.possible	1PL-catch-SBRD	1PL-fish

‘We’ll be able to catch our fish.’ (Adell 2019: 303)

The expression of desire exhibits similar variation. Languages such as Central Alaskan Yupik (Eskimo-Aleut, Example 11) and Meskwaki (Algic) utilize modal verbs for this function. Notably, in Meskwaki, the main event verb must be inflected for the future tense when expressing desiderative modality (Example 12).

- (11) Central Alaskan Yupik (Eskimo-Aleut)
qimugte-ka *qetunra-mnun*
 dog-ABS.1SG.PSR.SG.PSD son-ALL.1SG.PSR.SG.PSD
auluke-sqe-ssaaq-aqa
 take.care-ask-but-IND.1SG>3SG
 'I wanted my son to take care of my dog, but...' (Miyooka 2012: 1139)
- (12) Meskwaki (Algic)
ne-t-aka:wa:n-a:w-a=koh *wi:hne:w-a-k-i*
 1-EP-want-DIR-3-SG=certainly see.FUT-DIR-3-MODE
 'I do want to see him .' (Dahlstrom ND: 336)

Languages may also use dedicated desiderative morphemes or complex syntax. Huehuetla Tepehua (Totonacan) utilizes the suffix *-putun* on the main event verb (Example 13). In Tipai-Kumeyaay-Huerteno (Cochimi-Yuman), desire involves a clause-linking strategy where the verb *map* ('want') governs a complement clause marked by the nominalizer/demonstrative *pu* (Example 14). Onondaga (Iroquoian) employs the modal verb *adyhwejyo* ('want/need') with the benefactive suffix *-ni* (Example 15).

- (13) Huehuetla Tepehua (Totonacan)
wachuu-ch *waa* *k-tamakajun-putun* *?ani?*
 also-ALD FOC 1.SUBJ-stay-DES.IPFV here
 'I want to stay here, too.' (Kung 2007: 293)
- (14) Tipai-Kumeyaay-Huerteno (Cochimi-Yuman)
puu-ch *nya?wach* *neyiw-x-pu* *map*
 that.one-SUBJ we.SUBJ come.PL-IRR-DEM want
 'He wants us to come over.' (Miller 2001: 220)
- (15) Onondaga (Iroquoian)
na? *de-hon-adyhwejyo-ni-k* *ne?* *hy-a?shenih*
 ASRT DL-3M.NSG.P-want-BEN-HAB NMLZ 3M.PL.A-white.man
e? *na?* *a-hy-ad-yenaw-asd-a?* *nege ne?*

REP ASRT OPT-3M.PL.A-SREFL-hold.on-CAUS-PNC DEM NMLZ

ø-ɥhwɛjy-a?

NPF-land-NSF

‘What the white men want is for them to keep holding on to this land.’

(Woodbury 2018: 407)

Coreference constraints provide another axis of variation in desiderative expressions. In Ute-Southern Paiute (Uto-Aztecan), the matrix verb *ʔásti* (‘want’) takes a phrasal complement that varies based on subject identity. When the subjects are coreferential (Example 16a), the complement verb takes the irrealis marker and the nominalizer *-chi*. However, in different-subject contexts (Example 16b), the irrealis marker combines instead with the complementizer *-ku*.

(16) Ute-Southern Paiute (Uto-Aztecan)

a. *ʔáapa-chi wúuka-vaa-chi ʔástiʔi-y*
boy-SUBJ work-IRR-NMLZ want-IMM

‘The boy wants to work.’ (Givón 2011: 102)

b. *ʔáapa-chi tukuavi tuka-vaa-ku ʔásti-kya*
boy.OBJ meat.OBJ eat-IRR-COMP want-ANT

‘(S/he) wanted the boy to eat the meat.’ (Givón 2011: 217)

4.2 Semantic variation

Cross-linguistically, languages may employ distinct formal encodings for various modal sub-functions. For example, some languages distinguish between physical and mental ability (as in Lhomi, Sino-Tibetan; Vesalainen 2014), while others differentiate between types of necessity, such as strong, weak, and external obligation (as in Mosuo, Sino-Tibetan; Lidz 2010). Similarly, distinct forms may distinguish subtypes of desiderative modality, such as physical versus non-physical desire (as in Wadu Pumi, Sino-Tibetan; Daudey 2014). In our sample, however, such internal variation was found only within the domain of necessity. In Yuchi (isolate; Linn 2001), strong necessity (Example 17a) is expressed via the modal verb *thæ* (‘want’, ‘need’), while weak necessity (Example 17b) is encoded by the modal verb *te* (‘be able’) combined with a verbalizer. Furthermore, Yuchi employs a dedicated modal verb, *hetē* (‘must’), inflected in the future tense to express social or external obligation (Example 17c).

(17) Yuchi (isolate)

a. *pickett-he weda di-thæ*
Pickett-LOC 1SG.A.go 1SG.A-want
‘I need to go to Pickett.’ (Linn 2001: 493)

b. *peshē-ʔē-de kʔala kʔō te-ʔē*

- | | | | | |
|----|---|------------------------|------|--------------|
| | more-VBLZ-CMPL | thing | make | be.able-VBLZ |
| | 'I ought to work more.' (Linn 2001: 300) | | | |
| c. | <i>tsothishikʔōne-he</i> | <i>hē-dze-kʔā-thla</i> | | <i>hetē</i> |
| | doctor-LOC | SG.M.A-1SG.P-COM-go | | must.FUT |
| | 'He has to take me (to the doctor).' (Linn 2001: 301) | | | |

5 Data

To facilitate intra- and cross-linguistic analysis, each construction is annotated in a template format, enclosed in square brackets []. Each template includes the host (defined as the element designating the state of affairs) as well as the other elements minimally required to express the respective modal function. All elements are annotated according to their linear properties. In Example 18 (Ixil, Mayan; Example repeated from 10), the host within the template is the verb *txey* 'catch'. The morphosyntactic elements required to express the modal meaning of ability involve the tense marker *laʔ=*, the modal verb *utch*, and the subordinator *-t*. These minimally required forms were identified based on contrastive examples in the respective sources and annotated according to their morphosyntactic category (e.g., tense, aspect), function (e.g., dependent markers), and part of speech (e.g., verb, noun). Consequently, the template structure for Example 18 is [ASPECT=MOD.V1 HOST-SBRD]. Additionally, recurring marker types have been assigned distinct IDs (such as MOD.V1) to track marker identity across different constructions within the same language.

- (18) Ixil (Mayan) [ASPECT=MOD.V1 HOST-SBRD]
laʔ=utch qu-txey-t qu-txay
 PROS=be.possible 1PL-catch-SBRD 1PL-fish
 'We'll be able to catch our fish.' (Adell 2019: 303)

Intra- and interlinguistic variation across templates is annotated using a bottom-up approach; rather than using predefined variables, every minimal contrast is annotated pairwise. This approach, which aims to capture all observable linguistic variation rather than discarding it, is referred to as autotypology (Bickel & Nichols 2002). To capture this variation, we employ a construction-based multicategorization approach (Ivani & Bickel 2025). This means we do not rely on abstract language-based types; instead, we annotate and describe individual constructions as they emerge in the data. We then decompose formal and functional variables into binary variables that capture minimal differences between constructions. By comparing constructions pairwise and annotating new variables as they surface, we established a variation space consisting of 42

variables. Table 2 provides a coding excerpt of the annotated data. The full dataset is available on Zenodo.¹

Language	Template	TAM required?	Dependency marker required?	Can express ability?	Can express desire / volition	Modal has lexical meaning	Modal has lexical meaning 'know'?
Barbareño	MOD.MRK1-HOST	no	no	no	yes	no	no
Caddo	MOD.MRK1=MOOD1-	yes	no	no	no	no	no
Northeast Chiapas Zoque	MOD.V1-HOST-	yes	no	yes	no	no	yes
Sochiapam Chinantec	MOD.V3-SBRD	yes	yes	no	yes	yes	no

Table 2: Coding excerpt of the annotated data.

6 Constructions patterns

In the data, we find that most of the sampled languages employ both verbs and non-verbal markers in the relevant AOM constructions ($n = 13$). All languages in the sample use at least one verb, or verb-like form, in an AOM construction. In what follows, we first examine functional convergence: that is, which modal functions attested in the data—desire, necessity, ability, and related subtypes—are encoded in similar ways (Section 6.1). We then explore patterns of correlation between functions and specific forms, asking which forms are associated with which functions (Section 6.2). Given the widespread use of verbs in AOM constructions across the sample, we also consider whether particular verbs are associated with particular functions (Section 6.3). Finally, we report several qualitative observations concerning less common phenomena observed in the sample (Section 6.4).

6.1 Functions clustering

The data contain three main modal functions: desire, ability, and necessity, along with several necessity-related subtypes: weak necessity, strong necessity, and moral obligation. In this section, we explore which of these modal functions are formally encoded in similar ways and which are encoded differently.

We treat two functions as formally similar to the extent that the same constructions encode both, and as formally distinct to the extent that they are encoded

¹ DOI: [10.5281/zenodo.20671744](https://doi.org/10.5281/zenodo.20671744)

by different constructions. To quantify this across the sample, we build a binary matrix whose rows are modal functions and whose columns are language-specific constructions; each cell records whether the construction can encode that function. We then compute the pairwise Jaccard distance between rows, yielding a dissimilarity index for each pair of functions. The index approaches zero when two functions are largely encoded by the same constructions (high overlap) and approaches one when they are encoded by largely disjoint sets (low overlap). Figure 2 displays the resulting distance matrix.

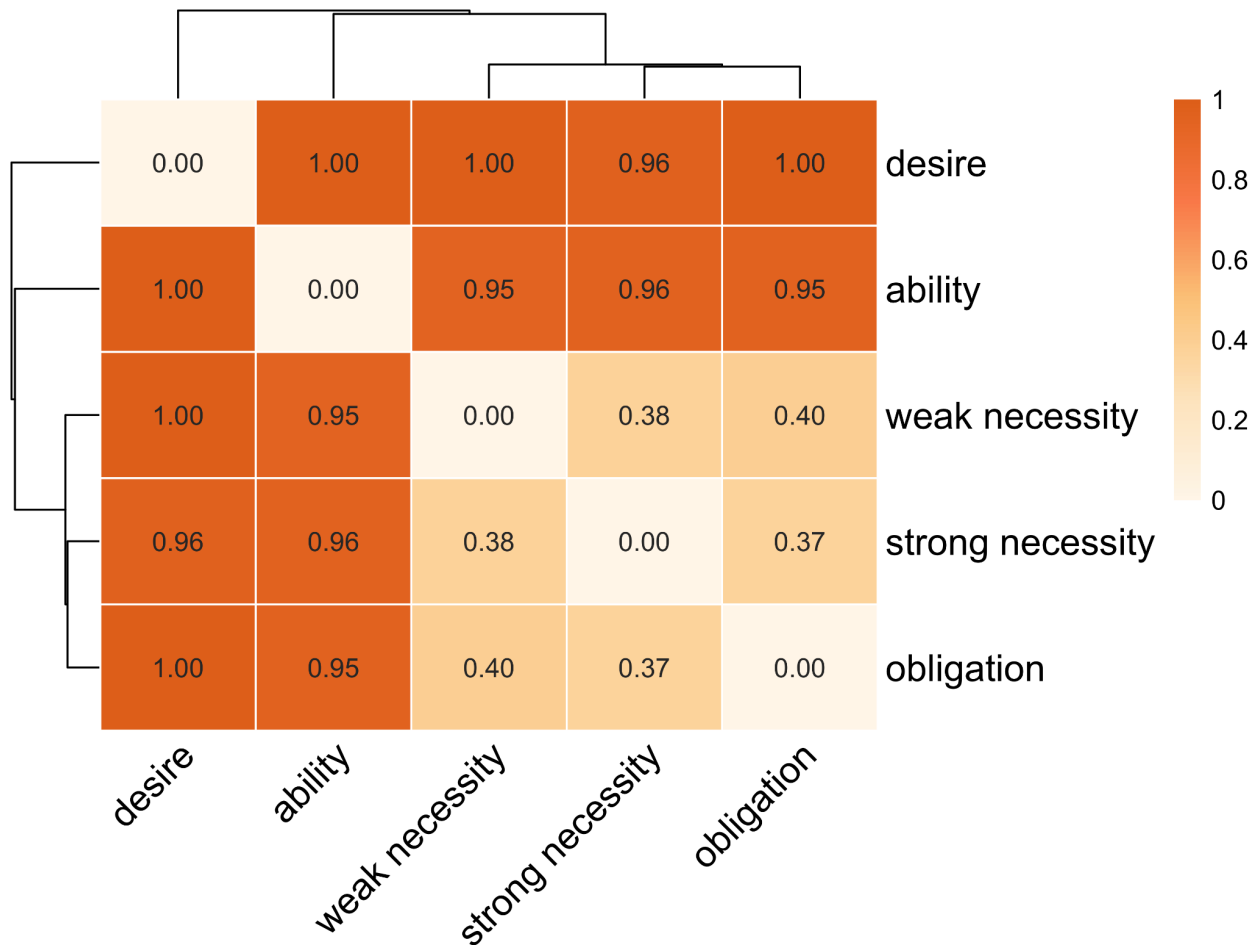


Figure 2: Encoding distance matrix for modal functions

The results reveal a clear cluster of necessity-related functions. Weak necessity, strong necessity, and moral obligation are relatively close to one another, with pairwise Jaccard distances ranging from 0.37 to 0.40. This indicates substantial overlap in the constructions used to encode these meanings. By contrast, desire and ability are sharply differentiated from the necessity-related functions and from each other. Desire has distances of 0.96–1.00 from the other functions, while ability has distances of 0.95–

1.00. These high values indicate little to no overlap in the constructions used to express these functions. Thus, the strongest pattern of formal convergence in the data is found among the necessity-related meanings, whereas desire and ability form distinct functional domains with little formal overlap either with necessity meanings or with one another.

6.2 Form patterns

We next explore whether specific formal properties are associated with particular modal functions — for example, whether the presence of tense marking, such as future marking, correlates with the expression of necessity. To investigate this, we fit a Bayesian multivariate logistic regression that predicts, for a given construction, the probability of expressing each of the five modal functions (strong necessity, weak necessity, obligation, desire, and ability) from its formal properties. Modeling the five functions jointly lets a single construction be associated with more than one function. We restricted the predictors to formal properties that vary across constructions in our sample — properties that are almost always present or almost always absent carry little information about function — and excluded those present in fewer or more than 15% of constructions. Since we are modeling individual constructions, we include a random intercept at the language level to absorb the within-language variation.

We summarize each form-function association as a marginal effect: the change in probability of a function when a formal property is present versus absent, reported with 90% (thin line) and 66% (thick line) credible intervals (Figure 3). Each panel shows one formal property, with the five modal functions as rows.² When the wider interval lies entirely to one side of zero, the posterior probability that the effect has that sign is at least 95% — to the right (blue), the property is positively associated with expressing the function; to the left (red), negatively. We treat these as credible associations. Where the 90% interval crosses zero but the 66% interval does not — corresponding to at least 83% posterior probability of a directional effect — we treat the evidence as suggestive only; weaker effects we treat as indistinguishable from null.

² While we recognize that we could just merge the modal functions of weak and strong necessity and obligation into one single modal function, for the sake of clarity and detail we report results for each modal function.

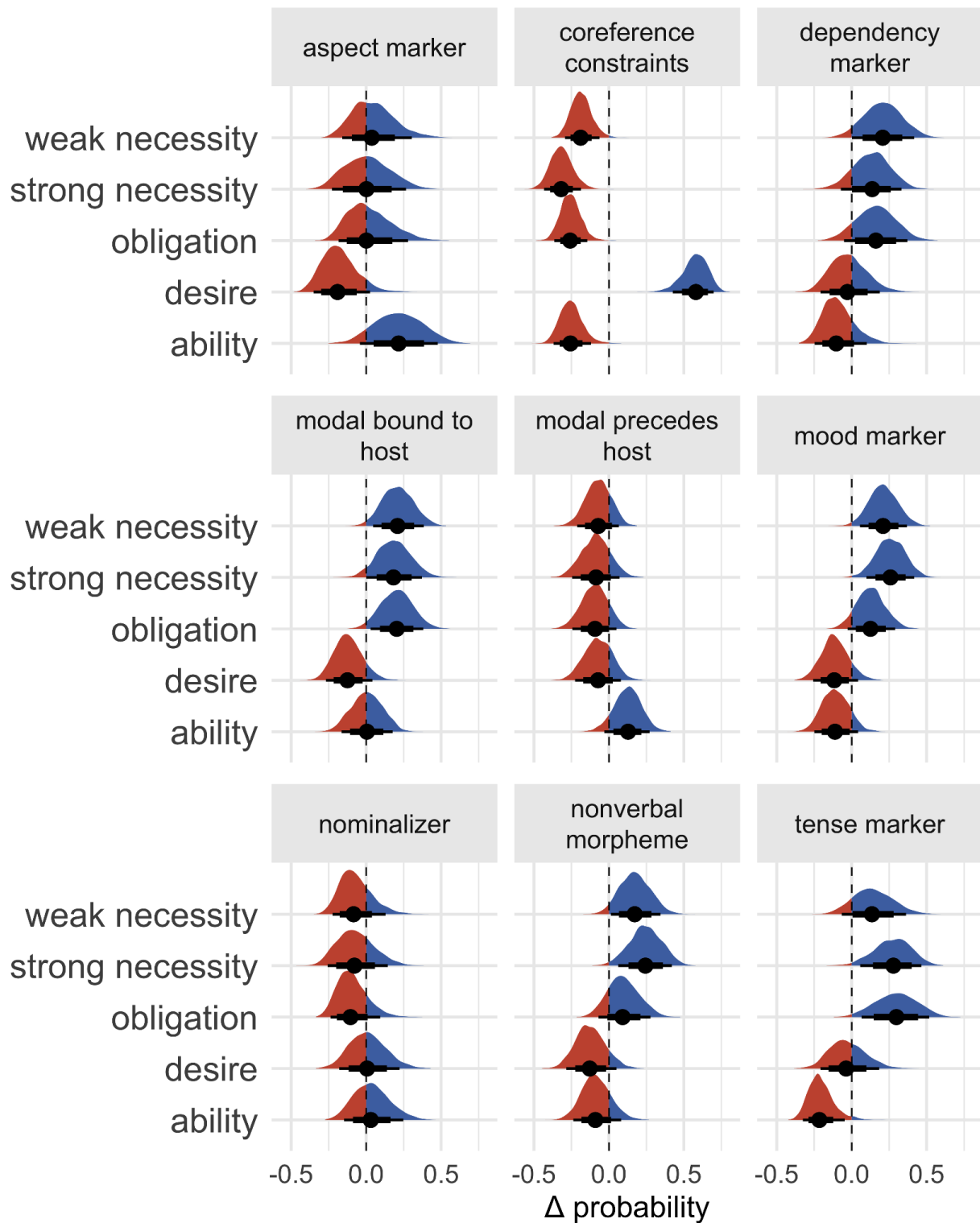


Figure 3: Marginal effects of formal properties on modal functions

Figure 3 shows several form-functions associations in the sample. First, tense marking is positively associated with strong necessity and obligation, while mood marking is positively associated with weak and strong necessity. These two verbal categories have a negative association with ability and desire. We also see weak evidence of association between tense marking and weak necessity, as well as between mood marking and obligation.

Aspect marking shows a different pattern. While it shows no interaction with the expression of desire, there is evidence of weak negative association with the expression of desire and weak positive association with the expression of ability. Non-verbal morphemes, such as modal particles, are positively associated with strong and weak necessity.

A similar pattern is found for parameters concerning the position and attachment of modal material. The positive association between necessity-related functions and the parameter “modal bound to host” suggests that, in these languages, markers expressing necessity are often morphologically dependent on the host verb. This is consistent with the broader association between necessity and bound or marker-like forms, including tense, mood, and nonverbal modal morphemes.

Finally, the positive association between coreference constraints and desire reflects the fact that desire is often expressed through complementation strategies in the languages of the sample. In such constructions, the interpretation of the embedded participant is typically constrained by the argument structure of the matrix predicate, giving rise to the observed association between desire and coreference-related properties.

6.3 Lexical verbs and modal functions

Given the widespread use of verbs in AOM constructions across the sample, we next examine whether particular lexical verbs are associated with particular modal functions. For this, we fit a Bayesian multivariate logistic regression that predicts, for each construction, the probability of expressing each of the modal functions (strong necessity, weak necessity, obligation, desire, and ability) based on the lexical meaning associated with the modal verb source (if the construction contains such a verb). As before, we include a random intercept for language. The marginal effects of presence of specific modal verbs on modal functions, with 90% (thin line) and 66% (thick line) credible intervals, are reported in Figure 4. When the wider interval line lies entirely to one side of zero, there is at least a 95% posterior probability that the presence of the specific modal verb the function in that direction — to the right (blue) that the verb is associated with expressing the function, to the left (red) the opposite. Where the 90% interval crosses zero but the 66% interval does not — corresponding to at least 83% posterior

probability of a directional effect — we treat the evidence as suggestive only; weaker effects we treat as indistinguishable from null.

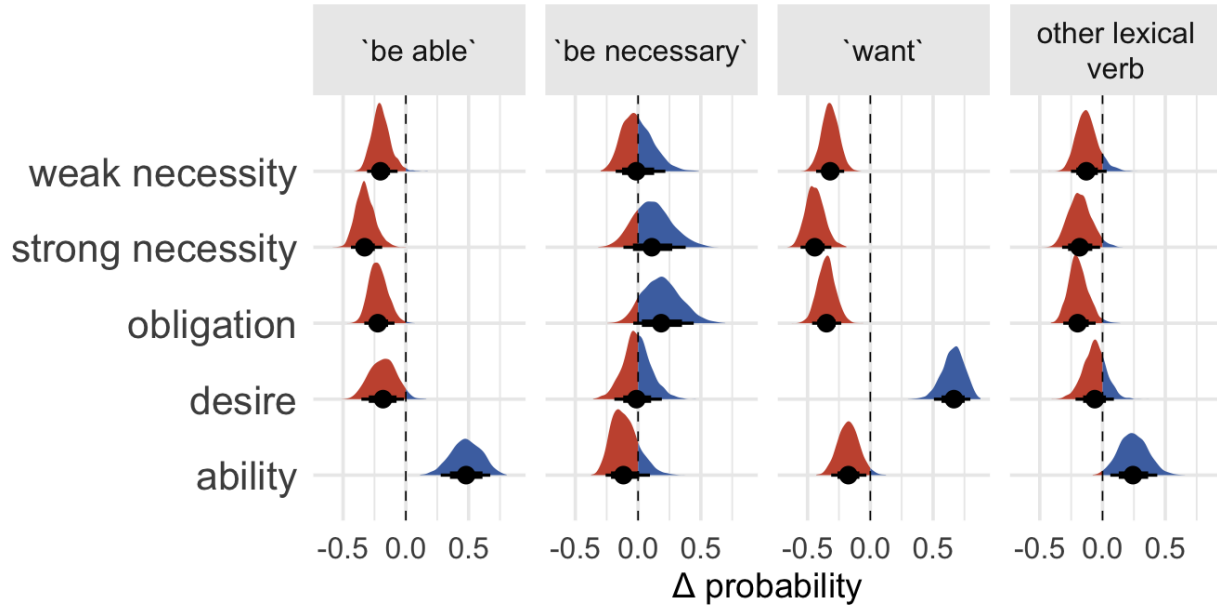


Figure 4: Marginal effects of modal verb semantics on modal functions

The distribution of lexical verbs and modal functions is relatively transparent. Verbs meaning ‘be able’ are positively associated with the expression of ability, as in Nuuchah-nulth (Wakashan; Example 19, repeated from 7). Similarly, verbs meaning ‘want’ are positively associated with the expression of desire, as in Meskwaki (Algic; Example 20, repeated from 12).

- (19) Nuuchah-nulth (Wakashan)
?u?um̥hi-matak-s *kʷis-<x>-aqs-i̯x̥*
 be.able.to-probably-1SG be.different-?-being.on.the.side-MOM
 ‘Maybe I can go across the strait.’ (Nakayama 2001: 115)
- (20) Meskwaki (Algic)
ne-t-aka:wa:n-a:w-a=koh *wi:hne:w-a-k-i*
 1-EP-want-DIR-3-SG=certainly see.FUT-DIR-3-MODE
 ‘I do want to see him.’ (Dahlstrom ND: 336)

The figure also shows that verbs classified as “other lexical verb”, that is, verbs whose lexical source is distinct from the semantically transparent types ‘be able’, ‘want’, and

‘be necessary’, are positively associated with ability. These cases are less straightforward semantically and will be discussed qualitatively in the following section.

6.4 Qualitative observations

An autotypological approach makes it possible to capture less common phenomena that would often be discarded in type-based typology. In this section, we report construction types and patterns of semantic variation that have limited occurrences in the sample. The examples discussed here occur only once in the data, but they are recoverable because our approach records all observable contrasts rather than reducing each language to a single type.

One less common construction used to express AOM meaning involves general extenders, that is, forms meaning ‘and such’, ‘and stuff like that’, or similar meanings. In Kiowa (Kiowa-Tanoan; Example 21, repeated from 6), weak necessity is encoded by suffixing the modal verb *dá* ‘must’ with the general extender *-àl*. The addition of the general extender softens the strong necessity meaning of the modal verb and yields a weak necessity reading.

- (21) Kiowa (Kiowa-Tanoan)
- | | | | | |
|--------------|-----------|--------------|------------|---------------|
| <i>dá-àl</i> | <i>ám</i> | <i>yí:dè</i> | <i>kôl</i> | <i>pj:gyá</i> |
| must-also | you | both | some | food |
- gyát-bó*
X.A>1PL.P>PL.OBJ-bring.IMP
‘You should also bring some food for us.’ (Watkins 1984: 222)

Another less common construction is found in Halkomelem (Salishan), where strong necessity is encoded through a complex construction involving the form ‘be impossible’ in combination with negation (Example 22). In this case, necessity is expressed indirectly, through the negation of impossibility.

- (22) Halkomelem (Salishan)
- | | | | |
|---------------|-----------|--------------------|----------------|
| <i>sk'wéy</i> | <i>mə</i> | <i>wə-ʔəwə-e:n</i> | <i>ném-e:n</i> |
| impossible | CERT | if-NEG-1SG | go-1SG |
- ‘I must go (lit. It is impossible that I do not go).’ (Suttles 2004: 132)

Less common patterns also arise in the lexical sources of modal verbs. Again in Halkomelem, ability can be expressed through verbs whose meanings are not transparently modal, including verbs meaning ‘be quick’ (*š'wém*, Example 23a) and ‘be enough’ (*s'łá'łəm*, Example 23b). These cases show how ability meanings may develop

from lexical sources that encode speed, sufficiency, or successful completion rather than ability in a narrow sense.

(23) Halkomelem (Salishan)

- a. *ha? ct ce? xíləs ?əý ǰwém kʷə*
 if 1PL FUT succeed and be.quick ART
s-qʷé-namər-ct
 NMLZ-penetrate-manage-1PL
 ‘If we succeed [in this plan], we can get through.’ (Suttles 2004: 457)
- b. *ni? wə-sǰáǰəm kʷə nə-s-ném*
 AUX EST-be.enough ART 1SG-NMLZ-go
wə-wéyəl-əs
 when-become.day-3.SBRD
 ‘I can go tomorrow.’ (Suttles 2004: 458)

Finally, in Onondaga (Iroquoian), the verb *eR* ‘think’ is used to express desire (Example 24). Although this pattern occurs only once in the present sample, the use of a cognition verb for desire is also attested crosslinguistically, for example in Mosuo (Sino-Tibetan; Lidz 2010) and Lhomi (Sino-Tibetan; Vesalainen 2016). This suggests that, while rare in our data, the extension from cognition to desire is not an isolated phenomenon.

(24) Onondaga (Iroquoian)

- do ga? n-wa?-w-adyet?-a? gəs*
 NDEF NDEF PTV-FACT-3N/Z.SG.A-number.of.times-PNC CST
h-eR-he? aa-h-ad-nasg-ŷ-?
 3M.SG.A-think-HAB OPT-3M.SG.A-pet-give-PNC
 ‘He frequently wants to give [away] his pet.’ (Woodbury 2018: 356)

7 Conclusion

This study has examined the morphosyntactic and functional encoding of agent-oriented modality in a maximally diverse sample of North American languages. Focusing on the expression of necessity, obligation, ability, and desire, we have shown that agent-oriented modality is not associated with a single constructional strategy. Rather, it is encoded through a wide range of morphosyntactic resources, including verbal morphology, tense-aspect-mood markers, modal verbs, verb-like elements, dedicated modal markers, and more complex periphrastic constructions. This variation confirms that agent-oriented modality is a cross-linguistically heterogeneous domain, both in

terms of the forms recruited for its expression and in terms of the semantic distinctions that individual languages encode.

At the functional level, the constructions in our sample cluster around three major modal domains: necessity, ability, and desire. Necessity includes meanings of obligation as well as strong and weak necessity, suggesting that social, circumstantial, and more general necessity meanings are often closely connected in the grammar. Ability forms a second major domain, frequently involving meanings related to capacity and successful realization. Desire constitutes a third domain, involving volitional meanings and, in some cases, meanings that approach intention or willingness. These clusters support the relevance of agent-oriented modality as a comparative domain, while also showing that the boundaries between its subtypes are not always discrete.

A central typological observation emerging from the study is that different modal functions show different associations with morphosyntactic realization. In particular, strong necessity and obligation meanings are often encoded by material associated with tense marking, while mood marking is positively associated with weak and strong necessity. This suggests that necessity is especially prone to being expressed through grammatical resources that already locate events with respect to reality status, temporal structure, or speaker commitment. Ability and desire, by contrast, show somewhat different constructional profiles, including lexical and predicate-based sources. These associations point to recurrent pathways by which languages recruit material from neighboring grammatical domains to encode agent-oriented modal meanings.

The analysis also highlights the importance of distinguishing between broad construction types and more fine-grained functional values. A single morphosyntactic strategy may encode more than one modal subtype, while a single modal function may be realized through several different constructional strategies. This many-to-many relation between form and function would be difficult to capture in a purely type-based typology. The templatic and autotypologizing approach used here makes it possible to preserve this complexity while still identifying cross-linguistic patterns. In this sense, the study contributes not only to the typology of modality, but also to methodological discussions about how to compare grammatical domains that are both semantically gradient and morphosyntactically diverse.

Several qualitative observations further illustrate the value of this approach. Less common constructions, including those involving negative predicates, general extenders, or lexical predicates such as ‘be quick’, ‘be enough’, or ‘be impossible’, show that agent-oriented modality may be expressed through resources that would be easily missed in a more restrictive coding scheme. These cases are typologically important precisely because they show how modal meanings may emerge from constructions that are not canonically modal. They also suggest that the diachronic sources of agent-oriented modality are broader than the best-known pathways involving verbs such as ‘want’, ‘know’, or ‘need’.

Abbreviations

1 ‘first person’, 2 ‘second person’, 3 ‘third person’, A ‘agent’, ABM ‘ablative-modalis’, ABS ‘absolutive’, ALD ‘already’, ALL ‘allative’, ANT ‘anterior’, ART ‘article’, ASRT ‘assertion’, AUX ‘auxiliary’, BEN ‘benefactive’, CAUS ‘causative’, CERT ‘certainty’, CIS ‘cislocative’, CMPL ‘completive’, COM ‘comitative’, COMP ‘complementizer’, CST ‘customary’, DECL ‘declarative’, DEM ‘demonstrative’, DEP ‘dependent’, DES ‘desiderative’, DIR ‘direct’, DL ‘dualic’, EP ‘epenthetic’, EST ‘established’, FACT ‘factual’, FOC ‘focus’, FUT ‘future’, GEN ‘genitive’, HAB ‘habitual’, HORT ‘hortative’, IA ‘intransitive animate’, IMM ‘immediate’, IMP ‘imperative’, IND ‘indicative’, IPFV ‘imperfective’, IRR ‘irrealis’, LNK ‘linking particle’, LOC ‘locative’, M ‘masculine’, MODE ‘mode’, MOM ‘momentaneous’, N ‘neuter’, NDEF ‘indefinite’, NEC ‘necessitative’, NEG ‘negation’, NMLZ ‘nominalizer’, NPF ‘noun prefix’, NSF ‘noun suffix’, NSG ‘non-singular’, OBJ ‘object’, OPT ‘optative’, P ‘patient’, PASS ‘passive’, PL ‘plural’, PNC ‘punctual’, PRON ‘pronoun’, PROS ‘prospective’, PSD ‘possessed’, PSR ‘possessor’, PTV ‘partitive’, REP ‘repetitive’, SBRD ‘subordinator’, SG ‘singular’, SR ‘switch reference’, SREFL ‘semireflexive’, SUBJ ‘subject’, VBLZ ‘verbalizer’, X ‘implied unspecified agent’, Z ‘feminine-zoic’

References

- Adell, Eric James. 2019. *The phonetics, phonology, and morphology of Chajul Ixil (Mayan)*. Austin, TX: University of Texas at Austin dissertation. <https://hdl.handle.net/2152/78376>
- Bickel, Balthasar & Johanna Nichols. 2002. Autotypologizing databases and their use in fieldwork. In Peter Austin, Helen Dry & Peter Wittenburg (eds.), *Proceedings of the international Irec workshop on resources and tools in field linguistics, Las Palmas, 26-27 May 2002*. Nijmegen: MPI for Psycholinguistics.
- Bybee, Joan, Revere Perkins & William Pagliuca. 1994. *The evolution of grammar: Tense, aspect and modality in the languages of the world*. Chicago, IL & London: The University of Chicago Press.
- Chafe, Wallace. 2018. *The Caddo language: A grammar, texts and dictionary based on materials collected by the author in Oklahoma between 1960 and 1970*. Petoskey, MI: Mundart Press.
- Dahlstrom, Amy. no date. *A grammar of Meskwaki, an Algonquian language*. Ms.
- Daudey, Henriëtte. 2014. *A grammar of Wadu Pumi*. Melbourne: LaTrobe University dissertation.
- Davidson, Matthew. 2002. *Studies in Wakashan (Nootkan) grammar*. Buffalo, NY: State University of New York at Buffalo dissertation.
- Faarlund, Jan Terje. 2012. *A grammar of Chiapas Zoque*. Oxford: Oxford University Press.

- Foris, David P. 2000. *A grammar of Sochiapan Chinantec* (vol. 135; Publications in Linguistics). Dallas, TX: The SIL International and the University of Texas at Arlington.
- Givón, Tom. 2011. *Ute reference grammar* (vol. 3; Culture and Language Use). Amsterdam: John Benjamins. <https://doi.org/10.1075/clu.3>
- Ivani, Jessica K. & Balthasar Bickel. 2025. Databases for comparative syntactic research. In Sjef Barbiers, Norbert Corver & Maria Polinsky (eds.), *The Cambridge handbook of comparative syntax* (Cambridge Handbooks in Language and Linguistics). 75–107. Cambridge University Press. <https://doi.org/10.1017/9781009179362.004>
- Kratzer, Angelika. 1981. The Notional Category of Modality. In Hans-Jürgen Eikmeyer & Hannes Rieser (eds.), *Words, worlds, and contexts: New approaches in word semantics* 38–74. Berlin & New York, NY: De Gruyter Mouton.
- Kratzer, Angelika. 2012. *Modals and conditionals: New and revised perspectives*. Oxford: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199234684.001.0001>
- Kung, Susan Smythe. 2007. *A descriptive grammar of Huehuetla Tepehua*. Austin, TX: University of Texas at Austin dissertation. <http://hdl.handle.net/2152/3137>
- Lidz, Liberty A. 2010. *A descriptive grammar of Yongning Na (Mosuo)*. Austin, TX: University of Texas at Austin dissertation. <http://hdl.handle.net/2152/ETD-UT-2010-12-2643>
- Linn, Mary Sarah. 2001. *A grammar of Euchee (Yuchi)*. Lawrence, KS: University of Kansas dissertation.
- Marlett, Stephen A. 1981. *The structure of Seri*. San Diego, CA: University of California San Diego dissertation.
- Martin, Jack B. 2011. *A grammar of Creek (Muskogee)* (Studies in the Anthropology of North American Indian Series). Lincoln, NE: University of Nebraska Press. <https://doi.org/10.2307/j.ctt1df4h3s>
- Melnar, Lynette R. 1998. *Caddo verb morphology*. Lincoln, NE: University of Chicago dissertation.
- Miller, Amy. 2001. *A grammar of Jamul Tiipay* (vol. 23; Mouton Grammar Library). Berlin & New York, NY: De Gruyter Mouton. <https://doi.org/10.1515/9783110864823>
- Miyaoka, Osahito. 2012. *A grammar of Central Alaskan Yupik (CAY)* (vol. 58; Mouton Grammar Library). Berlin & Boston, MA: De Gruyter Mouton. <https://doi.org/10.1515/9783110278576>
- Nakayama, Toshihide. 2001. *Nuuchahnulth (Nootka) morphosyntax* (vol. 134; University of California Publications in Linguistics). Berkeley & Los Angeles, CA: University of California Press.

- Park, Indrek. 2012. *A grammar of Hidatsa*. Bloomington, IN: Indiana University dissertation.
- Portner, Paul. 2009. *Modality* (Oxford Surveys in Semantics and Pragmatics). Oxford: Oxford University Press. <https://doi.org/10.1093/oso/9780199292424.001.0001>
- Portner, Paul. 2018. *Mood*. Oxford: Oxford University Press. <https://doi.org/10.1093/oso/9780199547524.001.0001>
- Rice, Keren. 1989. *A grammar of Slave* (vol. 5; Mouton Grammar Library). Berlin & New York, NY: De Gruyter Mouton. <https://doi.org/10.1515/9783110861822>
- Suttles, Wayne. 2004. *Musqueam reference grammar* (First Nations Languages). Vancouver, BC: University of British Columbia Press. <https://doi.org/10.59962/9780774850971>
- Vesalainen, Olavi. 2014. *A grammar sketch of Lhomi*. Ms.
- Walker, Neil Alexander. 2020. *A grammar of Southern Pomo*. Lincoln, NE: Nebraska University Press.
- Wash, Suzanne. 2001. *Adverbial clauses in Barbareño Chumash narrative discourse*. Santa Barbara, CA: University of California at Santa Barbara dissertation.
- Watkins, Laurel J. 1984. *A grammar of Kiowa* (Studies in the Anthropology of North American Indian Series). Lincoln, NE: University of Nebraska Press.
- Woodbury, Hanni. 2018. *A reference grammar of the Onondaga language*. Toronto, ON: University of Toronto Press. <https://doi.org/10.3138/9781487516093>